

Decision Trees

↳ Fundamental ML tool for both classification & regression tasks.

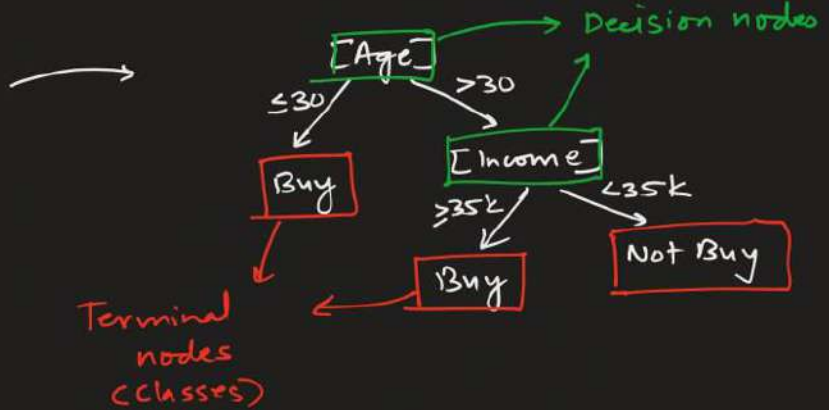
Structure of Decision Trees

- Nodes:- Each node represents a feature (or attribute) in the dataset.

- > Decision Nodes - Nodes that split into branches based on the value of the feature.

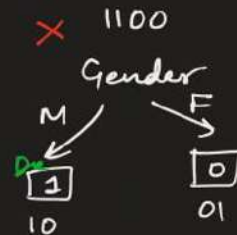
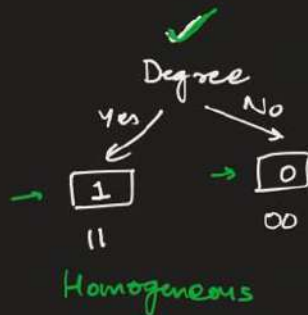
- > Leaf Nodes - Terminal nodes that provide the final output or prediction. Each leaf node corresponds to a class label in classification tasks or a value in regression tasks.

x		y
Age	Income	Buy/ Not Buy
35	40k	Yes
...	...	...



- **Splitting** :- The process of dividing a node into two or more sub-nodes based on a certain criteria. A good split will result in child nodes that are more homogeneous than the parent node.

	x	y
	Gender	(0/1)
→ Yes	M	1
→ No	M	0
→ Yes	F	1
→ No	F	0



• Depth of the Tree:- The longest path from the root node to a leaf node. Depth of the tree affects its performance; deeper trees may capture more detail but are also more prone to overfitting.

↳ model is too complex and captures noise in the data instead of the underlying distribution. This results in poor generalization to new data.

Underfitting

↳ A model is too simple to capture the underlying structure in the data. This can happen when the tree is NOT allowed to grow sufficiently.

Splitting criteria in Decision Trees

- Gini Index

The Gini index measures the impurity in the dataset.

$$\text{Gini}(D) = 1 - \sum_{i=1}^K p_i^2$$

D: Dataset

K: Number of classes

p_i : proportion of instances in class i

eg. Consider a dataset with following class distribution:-

class	# instances
A	10.
B	5.
C	5.

Calculate the Gini index for this dataset.

Sol.
$$\text{Gini}(D) = 1 - \sum_{i=1}^3 P_i^2$$

$$P_A = \frac{10}{20} = \frac{1}{2}, \quad P_B = \frac{5}{20} = \frac{1}{4}, \quad P_C = \frac{5}{20} = \frac{1}{4}$$

$$\begin{aligned} \text{Gini}(D) &= 1 - (P_A^2 + P_B^2 + P_C^2) \\ &= 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{4}\right)^2 - \left(\frac{1}{4}\right)^2 = \frac{5}{8} = \boxed{0.625} \end{aligned}$$

Gini index ranges from 0 to 1 $Gini \in [0, 1)$

> A Gini index of 0 indicates perfect purity (all instances belong to a single class)

> A Gini index of 1 indicates maximum impurity (instances are divided among classes)

eg. The maximum Gini index for binary classes = —

Sol. $Gini = 1 - P_A^2 - P_B^2$

$$P_B = 1 - P_A$$

$$= 1 - P_A^2 - (1 - P_A)^2$$

$$(P_A + P_B = 1)$$

$$\max(Gini) :- P_A = \frac{1}{2} = P_B$$

$$\max(Gini) = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = \boxed{\frac{1}{2}}$$

- Gini is minimized when all instances belong to single class
- Gini is maximized when there are equal number of instances in every class.

• Entropy

quantifies the randomness in the information being processed. It is a measure of uncertainty or impurity in a dataset.

$$H(D) = - \sum_{i=1}^K p_i \cdot \log_2(p_i)$$

D: Dataset

K: Number of classes

p_i : proportion of instances in class i

eg. Consider a dataset with following class distribution:-

Class	# instances
A	10.
B	5.
C	5.

Calculate the entropy of this dataset.

Sol.

$$\begin{aligned} H(D) &= - [P_A \cdot \log_2 P_A + P_B \cdot \log_2 P_B + P_C \cdot \log_2 P_C] \\ &= - \left[\frac{1}{2} \cdot \log_2 \frac{1}{2} + \frac{1}{4} \cdot \log_2 \frac{1}{4} + \frac{1}{4} \cdot \log_2 \frac{1}{4} \right] \\ &= - \left[-\frac{1}{2} - \frac{1}{2} - \frac{1}{2} \right] = \frac{3}{2} = \boxed{1.5} \end{aligned}$$

eg. What is the maximum and minimum entropy in a dataset with K classes?

Sol. Entropy is maximized when there are equal # instances in every class.

$$H(D) = - \sum_{i=1}^K P_i \log_2 P_i \quad P_i = \frac{1}{K}$$

$$H(D) = - K \times \frac{1}{K} \times \log_2 \frac{1}{K} = \boxed{\log_2 K}$$

Entropy is minimized when all instances belong to same class

$P_i = 1$, for all other classes $P_i = 0$

$$H(D) = - 1 \times \log_2 1 - 0 \dots = \boxed{0}$$

$$\boxed{0 \leq H(D) \leq \log_2 K}$$

- Information Gain

quantifies the reduction in entropy after a dataset is split on a particular feature.

$$IG(D, A) = H(D) - \sum_{v \in \text{values}(A)} \frac{|D_v|}{|D|} \cdot H(D_v)$$

D :- Dataset

A :- Feature on which we are splitting

D_v :- is the subset of D for the feature A with value v.

$|D_v|$:- is the number of instances in D_v

$|D|$:- number of instances in D

eg. Consider a dataset with following class distribution:-

class	# instances
A	10.
B	5.
C	5.

Assume that a split on a certain feature F with two values in the dataset results in two datasets with following composition:-

High:- $\{A:5, B:3, C:2\}$

Low:- $\{A:5, B:2, C:3\}$

Calculate the information gain if we split the dataset on feature F .

$$\text{Sol. } IG(D, F) = H(D) - \sum_{v \in \text{values}(F)} \frac{|D_v|}{|D|} \times H(D_v)$$

$$H(D) = - \left[\frac{10}{20} \times \log_2 \frac{10}{20} + \frac{5}{20} \times \log_2 \frac{5}{20} + \frac{5}{20} \times \log_2 \frac{5}{20} \right] = 1.5$$

$$H(D_{\text{high}}) = - \left[\frac{5}{10} \times \log_2 \frac{5}{10} + \frac{3}{10} \times \log_2 \frac{3}{10} + \frac{2}{10} \times \log_2 \frac{2}{10} \right] = 1.485$$

$$\underline{D_{\text{high}}} = [A:5, B:3, C:2] \quad |D_{\text{high}}| = 10$$

$$D_{\text{low}} = [A:5, B:2, C:3] \quad |D_{\text{low}}| = 10$$

$$H(D_{\text{low}}) = - \left[\frac{5}{10} \times \log_2 \frac{5}{10} + \frac{2}{10} \times \log_2 \frac{2}{10} + \frac{3}{10} \times \log_2 \frac{3}{10} \right] = 1.485$$

$$IG(D, F) = 1.5 - \left[\frac{10}{20} \times 1.485 + \frac{10}{20} \times 1.485 \right]$$

$$= \boxed{0.0145}$$

Choosing the Best Split:- The best feature for splitting is the one that provides the highest information Gain or the lowest gini index after the split.

eg: Consider a dataset containing 8 samples with the following class distribution:-

Class	Count
A	5
B	3

Calculate (i) Gini Index of the dataset
(ii) Entropy of the dataset

Sol: $Gini(D) = 1 - \sum_{i=1}^K p_i^2$ $P_A = \frac{5}{8}, P_B = \frac{3}{8}$

$$= 1 - \left[\left(\frac{5}{8} \right)^2 + \left(\frac{3}{8} \right)^2 \right] = \boxed{0.46875}$$

$$H(D) = - \sum_{i=1}^K p_i \log_2 p_i$$

$$= - \left[\frac{5}{8} \log_2 \frac{5}{8} + \frac{3}{8} \log_2 \frac{3}{8} \right] = \boxed{0.9544}$$

Details of ten international cricket games between two teams "Green" and "Blue" are given in Table C. This table consists of matches played on different pitches, across formats along with their winners. The attribute *Pitch* can take one of two values: spin-friendly (represented as *S*) or pace-friendly (represented as *F*). The attribute *Format* can take one of two values: one-day match (represented as *O*) or test match (represented as *T*).

A cricket organization would like to use the information given in Table C to develop a decision-tree model to predict outcomes of future games between these two teams.

To develop such a model, the computed $\text{InformationGain}(C, \text{Pitch})$ with respect to the Target is _____ (rounded off to two decimal places). ↑ ↑

Table C

Match Number	Pitch	Format	Winner (Target)
1	→ S	T	Green
2	→ S	T	Blue
3	F	O	Blue
4	→ S	O	Blue
5	F	T	Green
6	F	O	Blue
7	→ S	O	Green
8	F	T	Blue
9	F	O	Blue
10	→ S	O	Green

$$[G: 4, B: 6]$$



$$H(D) = - \left[\frac{4}{10} \times \log_2 \frac{4}{10} + \frac{6}{10} \times \log_2 \frac{6}{10} \right] = 0.971$$

$$H(D_S) = - \left[\frac{3}{5} \times \log_2 \frac{3}{5} + \frac{2}{5} \times \log_2 \frac{2}{5} \right] = 0.971$$

$$H(D_F) = - \left[\frac{1}{5} \times \log_2 \frac{1}{5} + \frac{4}{5} \times \log_2 \frac{4}{5} \right] = 0.722$$

$$\begin{aligned} \text{IG}(D, \text{Pitch}) &= 0.971 - \left[\frac{5}{10} \times 0.971 + \frac{5}{10} \times 0.722 \right] \\ &\approx \boxed{0.1245} \end{aligned}$$